

Article

Drug–Target Interaction Deep Learning-Based Model Identifies the Flavonoid Troxerutin as a Candidate TRPV1 Antagonist

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Abstract: Based on the advances made by artificial intelligence (AI) technologies in drug discovery, including target identification, hit molecule identification, and lead optimization, this study investigated natural compounds that could act as transient receptor potential vanilloid 1 (TRPV1) channel protein antagonists. Using a molecular transformer drug–target interaction (MT-DTI) model, troxerutin was predicted to be a TRPV1 antagonist at IC₅₀ 582.73 nM. In a TRPV1-overexpressing HEK293T cell line, we found that troxerutin antagonized the calcium influx induced by the TRPV1 agonist capsaicin in vitro. A structural modeling and docking experiment of troxerutin and human TRPV1 confirmed that troxerutin could be a TRPV1 antagonist. A small-scale clinical trial consisting of 29 participants was performed to examine the efficacy of troxerutin in humans. Compared to a vehicle lotion, both 1% and 10% *w/v* troxerutin lotions reduced skin irritation, as measured by skin redness induced by capsaicin, suggesting that troxerutin could ameliorate skin sensitivity in clinical practice. We concluded that troxerutin is a potential TRPV1 antagonist based on the deep learning MT-DTI model prediction. The present study provides a useful reference for target-based drug discovery using AI technology and may provide useful information for the integrated research field of AI technology and biology.

Keywords: cosmeceutical; drug repurposing; drug–target interaction; machine learning; MT-DTI; natural compound; sensitive skin; troxerutin; TRPV1



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1. Introduction

Transient receptor potential vanilloid 1 (TRPV1) is an ion channel protein expressed in sensory neurons. Activation of TRPV1 in response to stimuli, such as heat, capsaicin, or other chemicals, triggers an influx of cations such as calcium, magnesium, and sodium [1–3]. Capsaicin, a naturally occurring vanilloid, is a well-known TRPV1 agonist. TRPV1 is a major contributor to pain and is a potential target for analgesic agents [3]. It is also expressed in various non-neuronal organs, such as the bladder, lungs, cochlea, and epidermis, where TRPV1 activation is responsible for the pathological development of cystitis, asthma, hearing loss, and skin diseases, respectively [3,4]. The FDA approved Qutenza, a synthetic trans-capsaicin with a patch application system for postherpetic neuralgia [5]. Several TRPV1 antagonists, including ACD440 (NCT05416931), have been evaluated in preclinical and clinical studies for pain relief [6,7]. However, few studies have applied a machine learning (ML)-based approach for identifying candidate TRPV1 antagonists.

The use of artificial intelligence (AI) has increased in various areas, including the pharmaceutical industry [8,9]. ML technology—a major branch of AI—uses algorithms

that can be trained with known datasets to predict new outputs based on trained weights. In drug discovery, ML techniques can be used to predict the physicochemical properties, bioactivity, and toxicity of small-molecule bioactive compounds and are more cost-effective and rapid than conventional methods [8–11]. Using ML, the present study aimed to identify natural compounds or derivatives that could act as transient receptor potential vanilloid 1 (TRPV1) antagonists. Based on a molecular transformer (MT) drug–target interaction (DTI) model—previously used for predicting antiviral agents for SARS-CoV-2 [12,13]—we identified the natural flavonoid compound troxerutin as a potential TRPV1 antagonist from a target-based drug repurposing perspective. Furthermore, we investigated the bioactivities and potential of troxerutin in ameliorating skin sensitization and/or skin irritation.

2. Materials and Methods

2.1. MT-DTI Model and TRPV1 Inhibitor Hit Prediction

The model was built based on pre-trained MT chemical and convolutional neural network (CNN) protein amino sequence representations. It takes as input molecular sequences of drugs and protein sequences of targets and outputs affinity score for their interaction. The MT-DTI model was trained with chemical–protein pairs combined and curated from the Drug Target Common and BindingDB databases to predict binding affinity (K_D) and inhibitory bioactivity (half maximal inhibitory concentration [IC_{50}]) [14,15]. After retrieving MT-DTI sources from a GitHub repository (<https://github.com/deargen/mt-dti>, accessed on 14 April 2022), installation and implementation were performed following the guides provided in the repository. In brief, the Python Keras framework was used, and an 8-core TPU machine from Google Cloud was used to optimize the pre-training process. In the re-training of the main MT-DTI model, the pre-trained MT part was transferred to the main DTI for fine-tuning. To achieve optimal performance, the fine-tuning process was carried out using one of the most comprehensive DTI datasets, known as ChEMBL. ChEMBL is a highly regarded bioactivity database that has been curated to contain over 2.7 million compounds with bioactivity data against over 13,000 targets. These targets include a diverse range of proteins such as enzymes, ion channels, G protein-coupled receptors (GPCRs), and nuclear receptors. The inclusion of ChEMBL in the fine-tuning process ensures that the MT-DTI model is highly accurate and effective in predicting drug–target interactions.

For in silico screening, a catalog of 407,270 natural products was obtained from the Collection of Open Natural ProdUcTs (COCONUT) database [16]. Compounds with molecular weights (MW) in the range of 380–800 g/mol were used to predict their binding affinity for the target protein. Compounds in the COCONUT DB with MWs lower than 380 g/mol (100 of 407,270 compounds) were excluded due to their low annotation levels and natural compound likeness scores. Compounds with MWs higher than 800 g/mol were rare in the model training dataset; thus, an MW cut-off of <800 g/mol was set to reduce uncertainty. The drug–target interaction deep learning MT-DTI model was used to predict the K_D and IC_{50} of the sequence library [17]. The MT-DTI model predicted the effective K_D or IC_{50} values for the chemical (SMILES) and amino acid sequences (FASTA) of the target protein (Figure 1) with the TRPV1 amino acid sequence (UniProt ID Q8NER1) as input. The results were screened for compounds with $K_D < 500$ nM and classified into three groups based on the pharmacophore feature similarities of known TRPV1 antagonists as center compounds, including SB452533, JTS-653, or mavatrep, to increase the chance of successful hit identification (Table 1). Although IC_{50} values were also predicted, the results were not used for screening due to the relatively small size of the training dataset compared to that of the K_D -based DTI model. The pharmacophore features were calculated using an open-source cheminformatics RDkit (<https://www.rdkit.org>, accessed on 17 April 2022) and exported in 1575 bit-vector formats. Each bit refers to a particular combination of feature types (H-donor, H-acceptor, negative-ionizable, positive-ionizable, aromatic ring, and lumped-hydrophobe) and the hop distance between each feature type. The 1575-dimensional pharmacophore feature was analyzed in three dimensions via a

principal component analysis (PCA), and molecules with similar features were grouped via K-means clustering [18,19]. The similarity scores of predicted compounds to each center compound were calculated through clustering, and compounds were grouped based on the highest similarity score; compounds with similarity scores of less than 0.3 were not included in the grouping. From previous experiences of MT-DTI applied studies, we have experienced that pharmacophore feature-based similarity grouping or similarity grouping to known reference compounds was useful for sorting results of MT-DTI. Finally, for better tracking and purchasing of the screened compounds, natural compounds were canonicalized based on the chemical abstract service (CAS) number provided, annotated to their source organism, and their novel effects were considered.

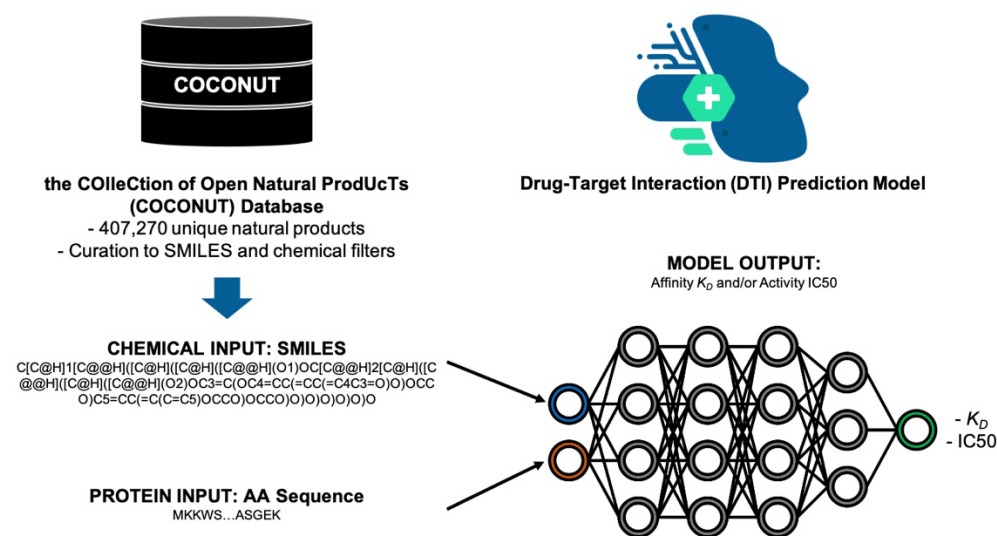


Figure 1. Overview of the affinity and bioactivity prediction process using the drug–target interaction (DTI) model.

Table 1. Information on known transient receptor potential vanilloid 1 (TRPV1) antagonists used as anchors for grouping predicted compounds with the molecular transformer drug–target interaction (MT-DTI) model.

Name	CAS No.	SMILES	Molecular Weight (g/mol)
SB452533	459429-39-1	<chem>O=C(NCCN(CC)C1=CC=CC(C)=C1)NC2=CC=CC=C2Br</chem>	376.29
JTS-653	942614-99-5	<chem>OC[C@@H]1N(C(N=C2)=CC=C2C)C3=CC=CC(C(NC(C=C4)=CN=C4OCC(F)(F)F)=O)=C3OC1</chem>	474.43
Mavatrep (JNJ-39439335)	956274-94-5	<chem>OC(C)(C)C1=CC=CC=C1C2=CC=C3N=C(/C=C/C4=CC=C(C(F)(F)F)C=C4)NC3=C2</chem>	422.44

2.2. Cell Culture and Measurement of Intracellular Calcium Influx in TRPV1-Overexpressing HEK293T Cells

A TRPV1-overexpressing HEK293T (TRPV1-HEK293T) cell line was purchased from Creative Biogene (New York, NY, USA) and cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 2 µg/mL puromycin and 10% fetal bovine serum (Gibco, Waltham, MA, USA) at 37 °C and 5% CO₂. At 80% confluence in a T75 flask, the cells were detached using trypsin EDTA (Gibco) and sub-cultured or seeded.

The experiment was performed as previously described [6,20–24]. Briefly, the TRPV1-HEK293T cells were treated with capsaicin (10 μ M) and troxerutin (1, 10, 50, and 100 μ g/mL) for 3 min, incubated with calcium-specific fluorescent dye Fluo-3/AM (5 μ M, Invitrogen, Waltham, MA, USA) and 0.1% Pluronic F-127 (Invitrogen) for 40 min, and the medium was washed using 1X NBS (normal bath solution, 140 mM NaCl, 5 mM KCl, 2 mM $\text{CaCl}_2/\text{EDTA}$, 0.5 mM MgCl_2 , 10 mM glucose, and 5.5 mM HEPES, pH 7.4). Intracellular calcium fluorescence responses were measured using a Leica DMi8 inverted microscope

(Leica Microsystems, Wetzlar, Germany) at excitation and emission wavelengths of 488 and 515 nm, respectively. Microscopic images were captured over 180 s at 1.5 s intervals. Intracellular calcium influx was determined using three independent experiments. Changes in intracellular calcium levels were expressed as an F/F_0 ratio, where F_0 is the initial fluorescence intensity, which was calculated using ImageJ (National Institutes of Health, Bethesda, MD, USA) with custom script. The dosage of 10 μ M capsaicin resulted in a fully activated response both in the present and a previous study [20].

2.3. Structural Modeling and Docking Analysis of Troxerutin

Due to the lack of a protein structure for human TRPV1 (hTRPV1), hTRPV1 (111-746) was generated using AlphaFold2 [25]. All protein and ligand preparation and scoring calculations were performed using the Schrödinger program (Maestro, version 13.1.141). The hTRPV1 protein structure was prepared using Schrödinger's Protein Preparation Wizard [26]. Purification and experiments of TRPV1 were performed in a buffer of pH 7.4 [27]. Hydrogen bond assignment was performed using PROPKA at pH 7.0, which can infer pK_a values based on empirical data and protonation followed by the method; then, a restrained minimization task was performed with default parameters. The ligand structures were ionized to possible states at a pH of 7 ± 2.0 . Receptor grids were generated from the TRPV1 agonist–capsaicin binding site [28]. Protein–ligand docking was performed using Glide [29,30]. Relative binding-free energy was calculated using Prime MM-GBSA, with the atomic length of amino acid residues to the docked ligand minimized to within 4.0 Å.

2.4. Clinical Evaluation of the Skin-Soothing Effect of Troxerutin

The recruitment of volunteers for the clinical study excluded individuals that were pregnant, receiving skin therapy, prone to skin irritation caused by 10% ethanol, or not reactive to capsaicin. The final cohort comprised 29 Korean females. Three formulations were tested: a vehicle lotion with no troxerutin, 10% troxerutin in vehicle lotion, and 1% troxerutin in vehicle lotion (Table 2). The participants were randomly divided into three groups, each tested with a different formulation (10 in the vehicle group, 10 in the 10% troxerutin group, and 9 in the 1% troxerutin group, with mean ages of 35.9, 33.1, and 34.2 years, respectively).

Table 2. Ingredients of lotions used in the clinical study. Troxerutin (w/v %) was incorporated by replacing an identical amount of water from each test lotion. The exact amounts of ingredients cannot be disclosed because of the information security policy of LG Household & Health Care.

Vehicle	10% Troxerutin	1% Troxerutin
Water	Water	Water
Pemulen TR2	Pemulen TR2	Pemulen TR2
Cetiol C5C	Cetiol C5C	Cetiol C5C
DPG-FG	DPG-FG	DPG-FG
Tris Amino Ultra PC	Tris Amino Ultra PC	Tris Amino Ultra PC
Activonol-6	Activonol-6	Activonol-6
Sepimax Zen	Sepimax Zen	Sepimax Zen
-	Troxerutin (10%)	Troxerutin (1%)

2.5. Immediate Soothing Effect Evaluation in Human Application

To measure the immediate soothing effect of troxerutin, an area of 1×1 cm was delineated on the forearm of each subject after acclimatization to 22 ± 2 °C and $45 \pm 5\%$ relative humidity for a minimum of 20 min. The skin redness of each test subject was measured using a Chromameter® (Konica Minolta, Singapore) before irritation with capsaicin (T0). After T0, participants applied a skin patch (IQ Ultra; Chemotechnique Diagnostics, Vellinge, Sweden) containing 40 μ L 0.1% w/v capsaicin solution (dissolved in 10% ethanol) to the forearm for 15 min to induce skin irritation (T1). Skin redness was measured again at 30 min after removing the patches (T2). After T2, lotions containing 10% troxerutin, 1%

troxerutin, or vehicle were applied to the forearm. After 30 min, the level of skin redness was measured to determine its calming effect (T3). Skin redness (a^*) was measured in triplicate, and the average value was used. We calculated the increase in redness due to capsaicin ($T2 - T0$) and decrease in redness due to the lotion use ($T3 - T0$) based on the skin redness before capsaicin treatment ($T0$). The skin-soothing rate was calculated using the following formula:

$$\text{skin soothing rate (\%)} = \frac{a^*_{T3} - a^*_{T0}}{a^*_{T2} - a^*_{T0}} \times 100$$

where a^*_{T0} represents basal skin redness, a^*_{T2} represents skin redness 30 min after capsaicin irritation, and a^*_{T3} represents skin redness 30 min after lotion application.

2.6. Statistical Analysis

Calcium influx data were analyzed using a one-way analysis of variance (ANOVA) with Tukey's post hoc test. Changes in skin redness before and after applying the products were analyzed using a paired Student's *t*-test. An unpaired Student's *t*-test was used to assess the differences between the vehicle and troxerutin 10% groups, and vehicle and troxerutin 1% groups. All statistical analyses were performed using Jamovi v.1.19 (The Jamovi Project, 2022; retrieved from <https://www.jamovi.org>, accessed on 21 October 2022) with significance set at $p < 0.05$.

3. Results

3.1. TRPV1 Antagonist Hit Prediction and Compound Selection

From the in silico screening of natural bioactive compounds from the COCONUT database using the MT-DTI model (Section 2.1), we identified 10 candidates (Table 3). Regarding a potential safety issue, antibiotic class compounds, such as karamomycin C, fredericamycin (fdm) A, fdm E, or an experimental cytotoxic anticancer agent amamistatin A, were excluded in order to proceed with wet experiments. Among these, troxerutin had a 0.333 pharmacophore–Tanimoto similarity score (maximum of 1.0) to the reference TRPV1 antagonist JTS-653 [24] and was selected based on the screening criteria (Section 2.1). To the best of our knowledge, troxerutin, a derivative of the bioflavonoid rutin, has not been previously reported as a TRPV1 antagonist.

Table 3. Candidate natural bioactive compounds predicted by the MT-DTI model.

COCONUT ID	Chemical Name	SMILES	TRPV1 Predicted K_D (nM)	TRPV1 Predicted IC_{50} (nM)	Group	Similarity Score
CNP0145155	Karamomycin C	<chem>COc1cc(C2=NC(C3SCC4C5SCC(C)(C(=O)N43)N5C)CS2)c(O)c2cccc12</chem>	43.65	397.84	SB 452533	0.344
CNP0303130	Amamistatin A	<chem>CCCCCCCC(OC(=O)C(CCCCN(O)C=O)NC(=O)c1nc(-c2cc(OC)ccc2O)oc1C)C(C)(C)C(=O)NC1CCCCN(O)C1=O</chem>	55.14	256.21	JTS-653	0.301
CNP0115741	Fdm E	<chem>CC=CC=Cc1cc2cc3c(c(O)c2c(=O)[nH]1)C1(CC3)C(=O)C(=O)c2c(c(O)c3c(O)c(OC)cc(O)c3c2O)C1=O</chem>	130.34	2011.20	JTS-653	0.301

Table 3. Cont.

COCONUT ID	Chemical Name	SMILES	TRPV1 Predicted K_D (nM)	TRPV1 Predicted IC_{50} (nM)	Group	Similarity Score
CNP0150586	Troxeutin	<chem>CC1OC(OCC2OC(Oc3c(-c4ccc(OCCO)c(OCCO)c4)oc4cc(OCCO)cc(O)c4c3=O)C(O)C(O)C2O)C(O)C(O)C1O</chem>	140.08	582.73	JTS-653	0.333
CNP0322312	Fdm A	<chem>CC=CC=Cc1cc2cc3c(c(O)c2c(=O)[nH]1)C1(CC3)C(=O)c2c(O)c3c(c(O)c2C1=O)C(=O)C(OC)=CC3=O</chem>	164.37	2727.29	JTS-653	0.359
CNP0273385	Plagiochin B	<chem>COc1ccc2c(c1)CCc1ccc(cc1)Oc1cc(cc(O)c1O)CCc1cc(O)ccc1-2</chem>	42.62	164.76	Mavatrep	0.324
CNP0237220	Dragmacidin D	<chem>CC(C1=CNC(N)N1)c1ccc(O)c2[nH]cc(-c3cnc(-c4c[nH]c5cc(Br)ccc45)c(=O)[nH]3)c12</chem>	11.83	29.34	No group	
CNP0295233	Dictazoline B	<chem>CN1C(=N)N(C)C2(C1=O)c1[nH]c3cc(Br)ccc3c1CC1(NC(N)=NC1=O)C2c1c[nH]c2cc(Br)ccc12</chem>	20.65	317.72	No group	
CNP0121123	Cryptophycin C	<chem>COc1ccc(CC2NC(=O)C=CCC(C(C)C=Cc3ccccc3)OC(=O)C(CC(C)C)OC(=O)C(C)CNC2=O)cc1Cl</chem>	32.90	217.69	No group	
CNP0295527	Eusynstyelamide D	<chem>NCCCCNC(=O)C1(O)C(c2c[nH]c3cc(Br)ccc23)C(O)(Cc2c[nH]c3cc(Br)ccc23)C(=O)N1CCCCN</chem>	142.69	278.27	No group	

3.2. Antagonism of Troxeutin on Calcium Influx in TRPV1-Overexpressing HEK293T Cells

We evaluated the effect of troxeutin on TRPV1 activation in TRPV1-HEK293T cells and found that 10 μ M capsaicin significantly increased calcium levels in the control group, and cotreatment with 100 μ g/mL troxeutin inhibited TRPV1 activation ($p < 0.001$, Figure 2). This suggests that troxeutin is a target-dependent antagonist of TRPV1.

3.3. Structural Modeling and Docking Poses of Troxeutin Compared to JTS-653

Ligands in hTRPV1–troxeutin and –JTS-653 were docked at binding sites between S4 and S5 of hTRPV1 (Figure 3) [28]. The docking scores of troxeutin and JTS-653 were -6.234 and -8.010 , and their mmGBSA ΔG binding free energies were -46.90 and -69.30 kJ/mol, respectively (Table 4). The tendency of the docking and mmGBSA scores was matched with the activity data of the ligands where the reported bioactivity of JTS-653 was IC_{50} 0.236 nM with 30 nM capsaicin [24], and troxeutin would have an IC_{50} in approximately a two-digit micromolar potency range with 10 μ M capsaicin. Interestingly, the ligands docked into the binding pocket without interacting with R557—a key residue for agonists [31]. Troxeutin has been predicted to form a hydrogen bond interaction with E570 (GLH570, protonated E570)—a key interaction for antagonists [27]—and had a hydrophobic interaction with I569 and A566 (Figure 4a). JTS-653 also formed hydrophobic interactions with I569 and A566 (Figure 4b). In contrast, hydrogen bonding with the E570 of troxeutin was not observed in

JTS-653. Instead, JTS-653 formed a hydrogen bond interaction with T550 and a π - π -stack interaction with F543. This enabled the stable binding of JTS-653 without a hydrogen bond interaction with T550. However, these results are inconclusive because we used the predicted structure of hTRPV1 in the absence of its crystal structure.

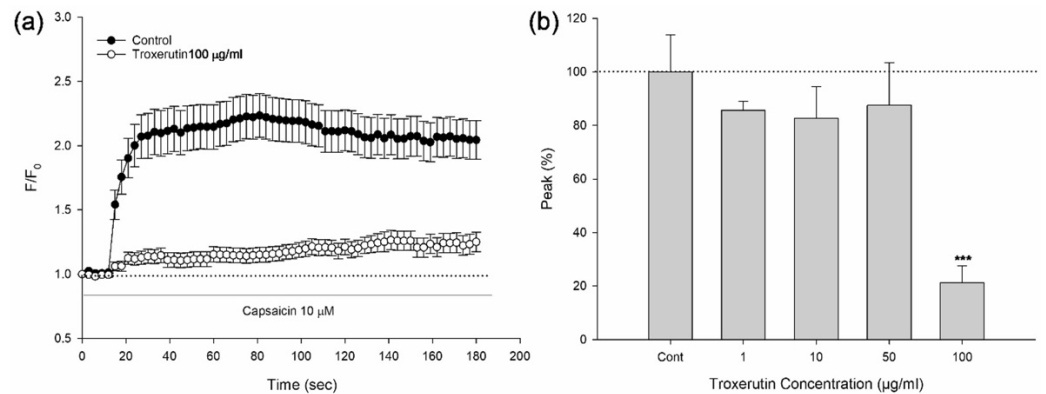


Figure 2. Capsaicin-induced calcium influx in TRPV1-overexpressing HEK293T cells. (a) Calcium influx induced by 10 μM capsaicin in the absence (control) or presence of 100 $\mu\text{g/ml}$ troxerutin for 180 s. (b) Summary of peak calcium influx (F/F_0) induced by treatment with 10 μM capsaicin and troxerutin (1, 10, 50, and 100 $\mu\text{g/ml}$). *** $p < 0.001$, ANOVA with Tukey's post hoc test.

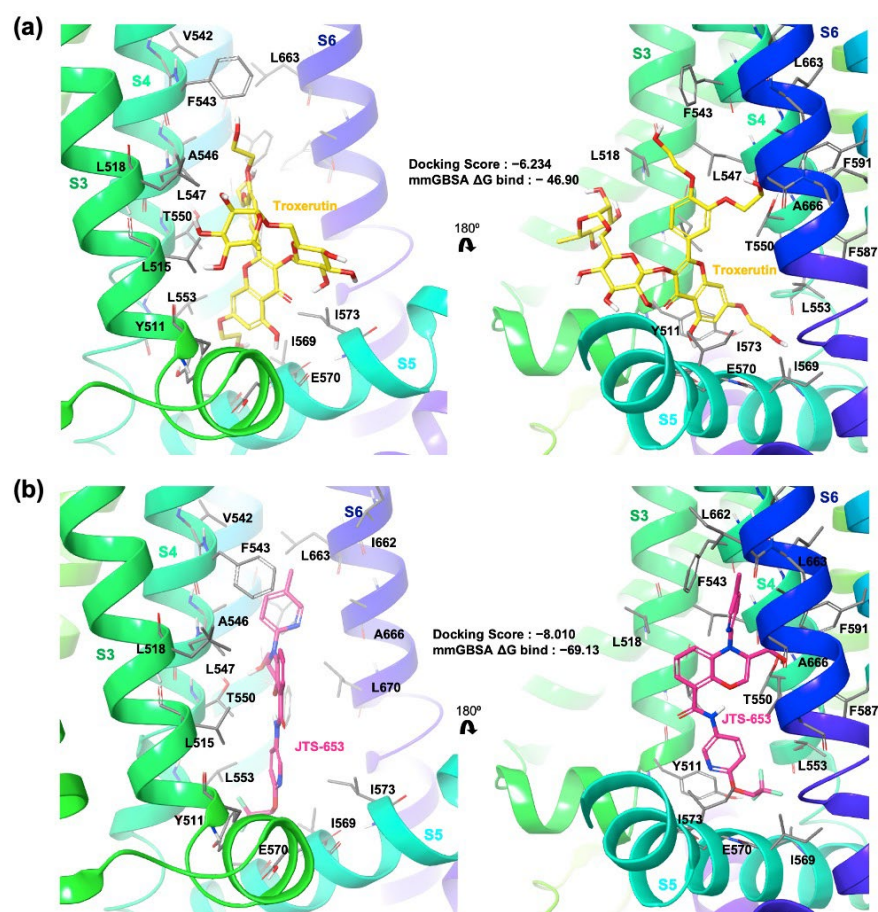
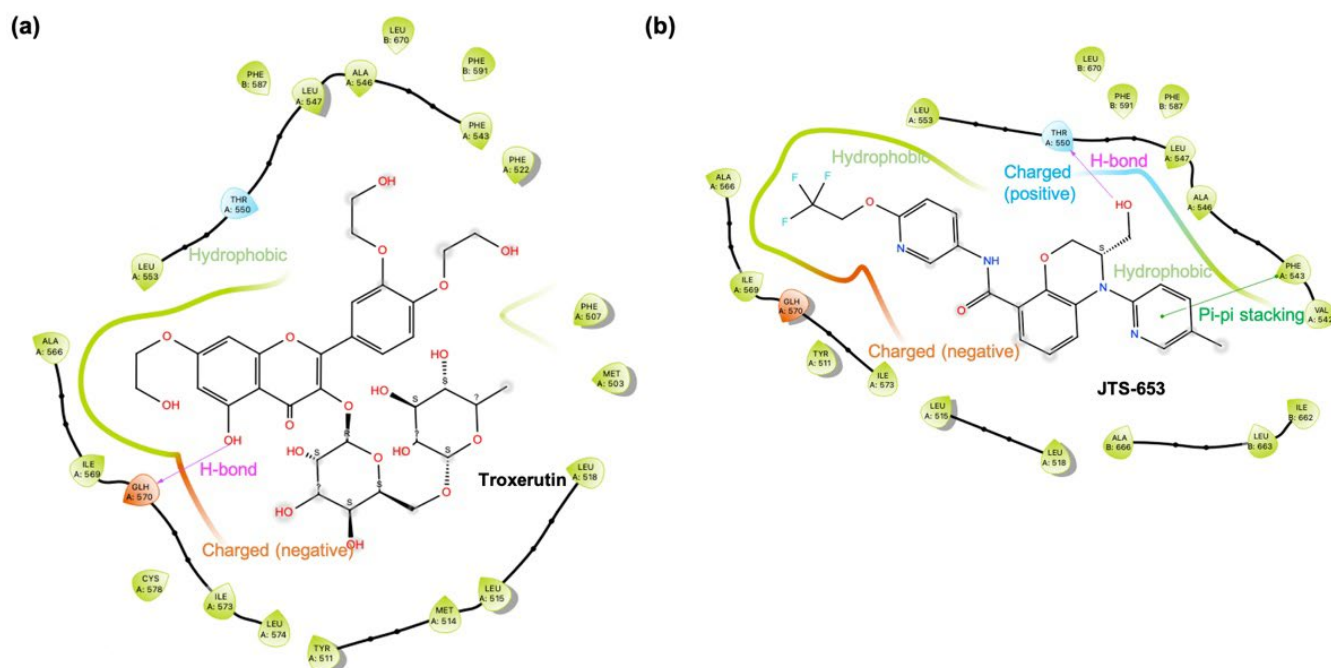


Figure 3. Structural characteristics of the troxerutin- and JTS-653-hTRPV1 complexes. (a) Prediction of the binding mode of troxerutin (yellow) and (b) JTS-653 (magenta) to hTRPV1. Docking and mmGBSA scores of each pair are annotated.

Table 4. Troxerutin docking and mmGBSA scores compared to those of JTS-653.

Chemical Name	Docking Score	mmGBSA ΔG Binding Free Energy
Troxerutin	−6.234	−46.90
JTS-653	−8.010	−69.13

**Figure 4.** Two-dimensional protein–ligand diagrams of molecular interactions between key residues of hTRPV1 and (a) troxerutin or (b) JTS-653. GLH: protonated glutamic acid residue.

3.4. Soothing Effect of Troxerutin on Skin Irritation Stimulated by Capsaicin

The increased levels of skin redness ($T2 - T0$) caused by capsaicin were similar among all three groups, and statistical significance was not observed among the groups upon capsaicin treatment (Figure 5). In contrast, statistically significant reductions in skin redness ($T3 - T0$) were observed in the 10% and 1% troxerutin groups. Although not statistically significant, an increase in skin redness was observed in the vehicle group from before to after the application of the lotion. The skin-soothing rates were 111.3%, 84.3%, and 87.9% in the vehicle, 10% in the troxerutin group, and 1% in the troxerutin group, respectively. Compared to the vehicle group, skin redness decreased in both the 10% troxerutin and 1% troxerutin groups ($p < 0.05$).

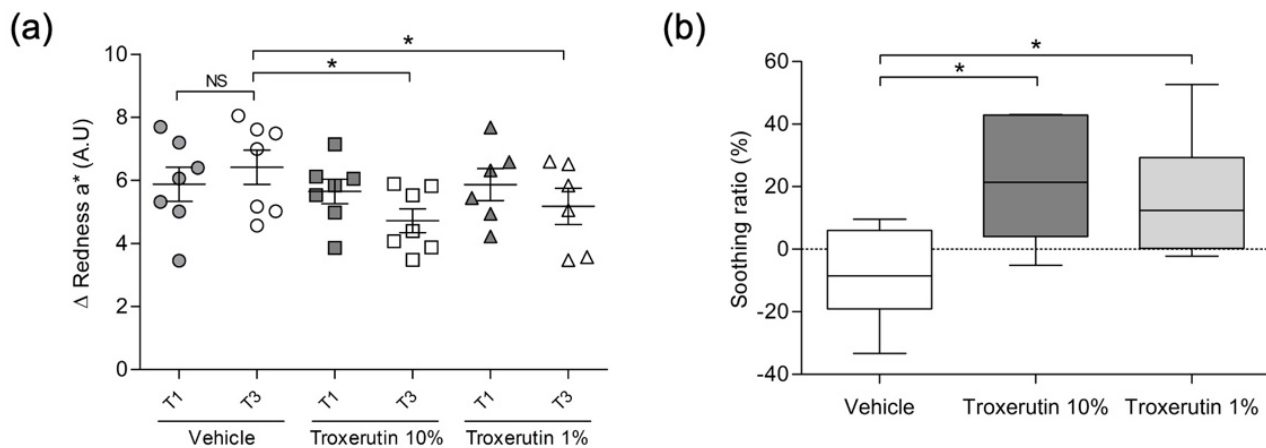


Figure 5. Immediate soothing effect of troxerutin after stimulation with 0.1% *w/v* capsaicin. (a) Variation in skin redness immediately after treatment with troxerutin lotion. Data are presented as the mean $\pm$ standard deviation (SD). NS, not significant; * $p < 0.05$ compared to the vehicle group at T3. (b) Soothing rate following application of lotions. Data are presented as the mean $\pm$ standard error (SE). * $p < 0.05$ compared with the vehicle group.

4. Discussion

Issues related to sensitive skin are increasing due to environmental and lifestyle changes. The associated symptoms, such as itching, stinging, burning, and pain, can severely reduce the quality of life. Therefore, it is important that effective cosmeceuticals are developed to ameliorate the symptoms of sensitive skin. In the present study, TRPV1—a receptor that is activated in sensitive skin—was used as a target to discover natural bioactive compounds that could reduce irritation by capsaicin—a TRPV1 agonist.

TRPV1 is a cation channel that promotes calcium influx when activated; thus, we confirmed the activation and inhibition of TRPV1 through calcium imaging. Increased calcium levels following treatment with 10 μ M capsaicin proved that TRPV1 was activated without abnormality in the cell model. Troxerutin significantly blocked capsaicin-induced TRPV1 activation, suggesting that troxerutin could be a potential antagonist that inhibits the effects of stimuli, such as chemical contact, itching, and pain, on skin sensitivity.

Troxerutin is a flavonoid found in a variety of plants, including the Japanese pagoda tree *Sophora japonica*. Troxerutin exhibits various beneficial activities, such as free radical scavenging [32], anti-arrhythmogenic and anti-inflammatory effects against ischemia/reperfusion injury of diabetic myocardium [33], blood pressure and lipid metabolism regulation [34], and elastase inhibition resulting in anti-inflammatory effects [35]. Although previous reports support the phenotypic efficacy of troxerutin against certain diseases, the target protein-based pharmacological interaction remains unclear due to its multimodal effects [36]. To our knowledge, the present study is the first to identify the antagonistic effects of troxerutin against TRPV1 using ML, which could promote target-based mode-of-action studies of troxerutin. Troxerutin has been widely studied in drug development owing to its oral bioavailability and low toxicity (rat LD_{50} = 27 160 mg/kg). However, previous clinical studies reported a maximum plasma concentration (C_{max}) of 2931 ± 1018 pg/mL following oral administration of 300 mg troxerutin, which makes it challenging to develop troxerutin as an orally administered drug or supplement against a wide range of indications [14,37]. Although the oral administration route may be insufficient, a study using a stripping method with troxerutin showed an excellent skin permeation rate of 80%, suggesting that troxerutin possesses sufficient competitiveness as a cosmeceutical compound, consistent with our hypothesis [32].

The interaction between troxerutin and TRPV1 seems to bear an effective concentration between 50 μ g/mL and 100 μ g/mL (67–134 μ M) in the presence of 10 μ M capsaicin. Based on the K_D and IC_{50} values estimated in the DTI model, future studies could potentially measure K_D and IC_{50} values experimentally via surface plasmon resonance, isothermal

titration calorimetry, microscale thermophoresis, or patch-clamp assays to accurately examine the relationship between predictions and wet-experiment results. The TRPV1 response inhibition rate of troxerutin was ~80% at 134 μ M; however, the predicted values of the DTI model were not directly correlated to the experimental values. The accuracy of the ML model could potentially be improved by additional drug–target data feed-in. In the structural modeling experiment, troxerutin and JTS-653 were found to have similar docking positions at the binding pocket of hTRPV1. Neither compound formed hydrogen bonds with R557—a key residue for agonists—but both interacted with GLH570 (protonated E570)—a key residue for antagonists. In addition, the compounds were predicted to have hydrophobic interactions with I569 and A566 in the S4–S5 linker, which could enhance binding stability [27]. These results not only provide the binding pose of troxerutin on TRPV1 but also suggest strategies for further design rationales of TRPV1 antagonists. Additionally, structural modeling of the human TRPV1–troxerutin complex also supports the prediction of the DTI model as well as the *in vitro* target-dependent antagonistic activity of troxerutin observed in the TRPV1-HEK293T model. Finally, the small-scale clinical application study showed detectable and statistically significant skin-soothing rates associated with the troxerutin lotions, consistent with the *in vitro* results. However, due to the enrolment schedule, these clinical results are gender specific and require further confirmation of generalizability to the male population.

AI is a crucial tool for rapid drug discovery, with many big pharmaceutical companies collaborating with AI drug discovery companies to enhance the drug development process [9,38]. In the drug discovery process, it is also essential that the activity, safety, and molecular signature of the compound be determined to select an effective bioactive compound for cosmeceutical usage. Therefore, AI tools are integral to establishing effective strategies for the discovery of cosmeceutical or cosmetic bioactive compounds. The results of the present study showed that the efficacy of troxerutin needs to be improved before considering its clinical application. Here, AI tools could also be applied to estimate the efficacy of various molecular modifications and design optimized compounds [39,40].

To conclude, the present study successfully demonstrated the AI-aided target-based discovery of natural compounds that could inhibit TRPV1 activation and potentially ameliorate skin sensitivity/irritation. The process and strategy applied in the present study could effectively accelerate cosmeceutical drug discovery.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/app13095617/s1>, PDB files of JTS-653 and troxerutin docked to predicted human TRPV1 protein.

Author Contributions: H.Y. performed the structural modeling and docking studies and participated in the writing of the manuscript. J.L., Y.J.L. and T.-Y.K. conducted the *in vitro* and human application studies, analyzed the results, and participated in the writing and editing of the manuscript. G.B. and Y.-h.K. performed post-processing of the MT-DTI prediction results and data refinement and participated in writing the manuscript. B.R.B. and M.-S.K. conceived and designed the experiments and logic of the present study and participated in the writing, editing, and correspondence of the manuscript. J.-M.L., S.-W.P. and Y.-S.S. contributed to supervision and project administration. All authors agree to be personally accountable for their own contributions and for ensuring that questions related to the accuracy or integrity of any part of the work, even ones in which an author was not personally involved, are appropriately investigated, resolved, and documented in the literature. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: Sources and information on MT-DTI are uploaded on the GitHub page (<https://github.com/deargen/mt-dti>, accessed on 14 April 2022). PDB files of the docking model of hTRPV1 with troxerutin or JTS-653 are provided as Supplementary Material of the present study.

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